EXPERIMENT 10

BI Deployment, Monitoring and Backup & Restore using Microsoft Power BI

AIM

To understand and design Business Intelligence deployment, monitoring, backup, and recovery strategies using Microsoft Power BI for ensuring availability, reliability, and continuity of BI solutions.

TOOLS / SOFTWARE REQUIRED

- Microsoft Power BI Desktop
- Microsoft Power BI Service
- Global Superstore Dataset.xlsx
- Microsoft Excel
- Microsoft PowerPoint (Optional)

DATASET AND POWER BI FILE USED

Global_Superstore_2018.xlsx loaded into EXP_PBI - 10.pbix: star schema with Fact_Sales, Dim_Customer, Dim_Product, Dim_Region, Dim_Date, plus the tables Business Scorecard and Backup Schedule. Pages: Dashboard, Relational Report, Matrix Report, Comparative Analysis, Executive Dashboard, Business Scorecard. Data: 51,290 order lines, sales $12,642,502, profit $1,467,457, profit margin 11.6%.

THEORY

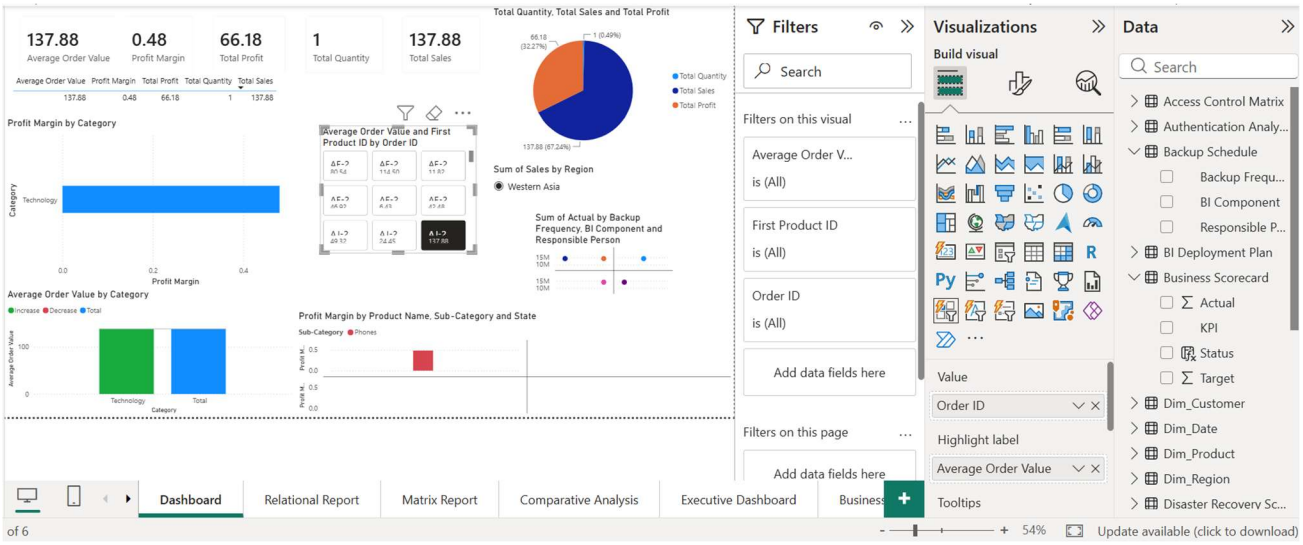
BI systems must be deployed, monitored, maintained and protected to ensure continuous business operations. Deployment makes reports, dashboards and analytical solutions available to end users. Monitoring tracks performance, usage, refresh status and operational health. Backup and restore protect BI assets from accidental deletion, corruption or system failure. Good deployment and maintenance strategies improve reliability, security and business continuity.

BUSINESS SCENARIO

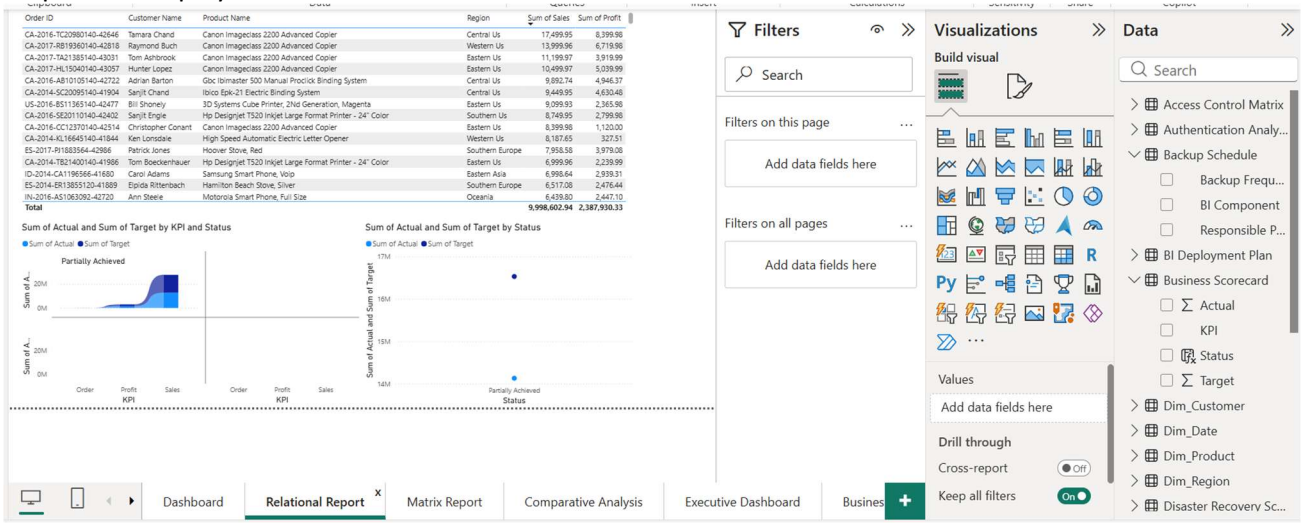
Global Superstore has developed reports, dashboards, analytical solutions, security policies and automated reporting. Management now wants to deploy the BI solution across the organization, monitor usage and performance, and establish backup and recovery procedures. As BI Solution Architect, a complete deployment and maintenance strategy is designed.

CONDUCT - PART A: BI DEPLOYMENT PLANNING

Step 1-2: BI assets from previous experiments



Step 3-5: BI Deployment Plan



Department	Intended users	Content deployed
Executive	CEO, Directors	Executive Dashboard, Business Scorecard
Sales	Sales Managers	Dashboard, Comparative Analysis, Relational Report
Finance	Finance Managers	Dashboard, Matrix Report, Business Scorecard
Regional Management	Regional Managers	Dashboard, Matrix Report (own region via RLS)

Deployment objectives and expected outcomes:

- Make one trusted version of the reports available to all departments (single source of truth).

- Reduce manual reporting effort through automated refresh and subscriptions.
- Keep data secure using Viewer access, Apps and RLS.
- Target availability of 99% during business hours and refresh completed before 06:00.

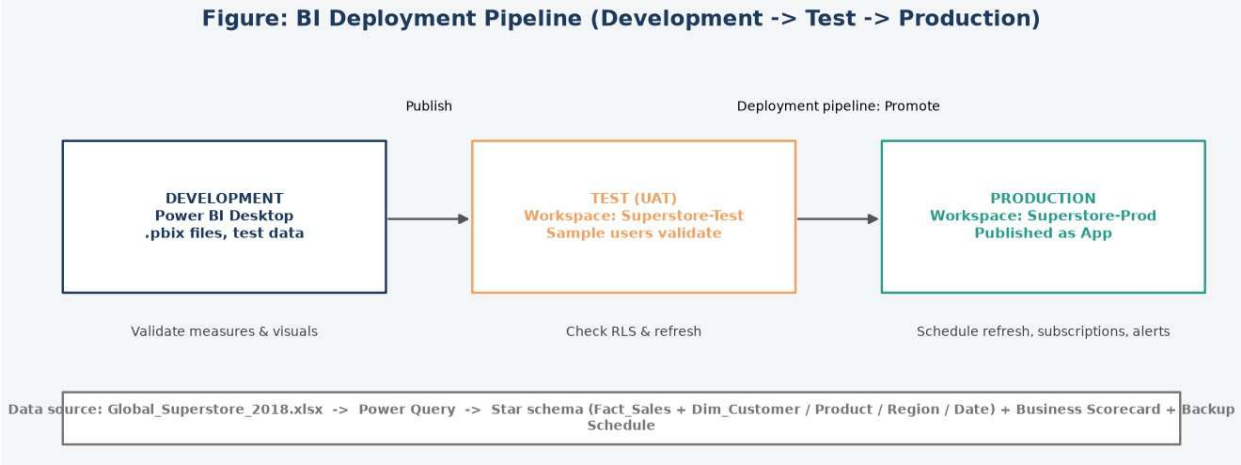
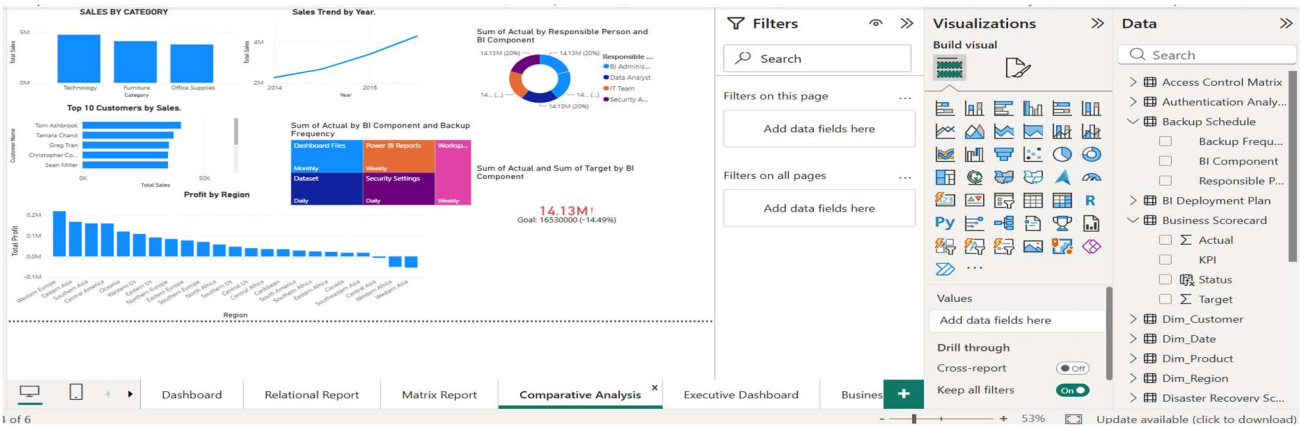


Figure 1: Deployment pipeline (Development, Test, Production).





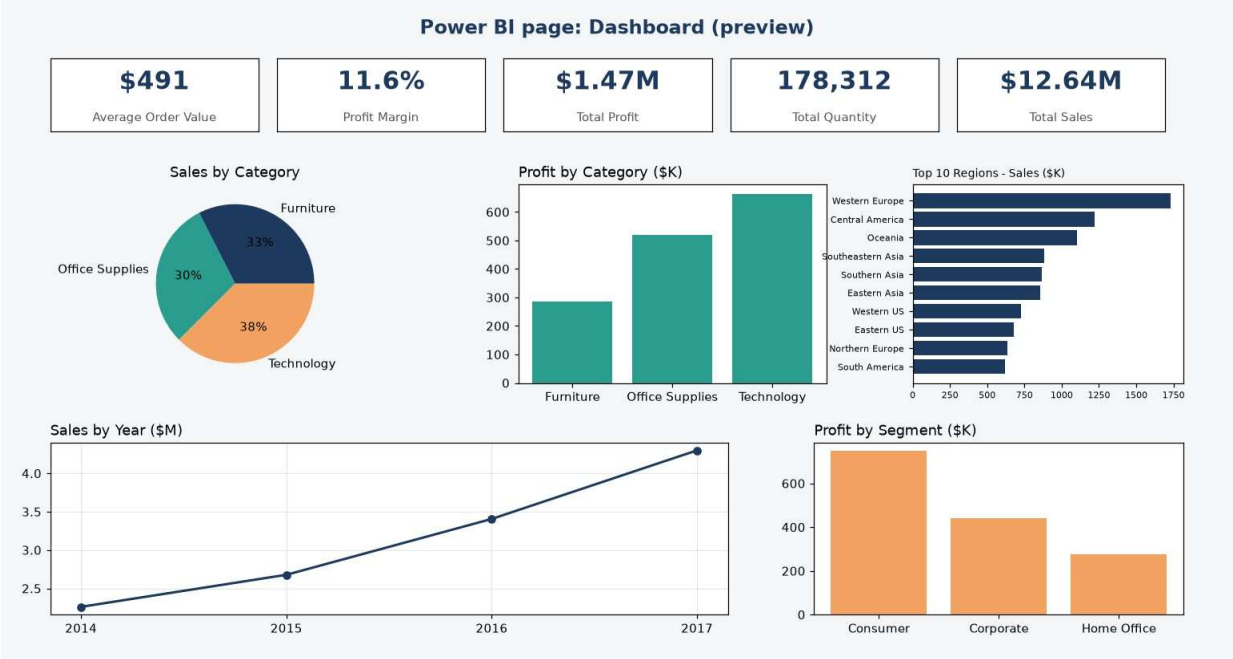


Figure 2: Dashboard page of the deployed report (preview; replace with Power BI screenshot).

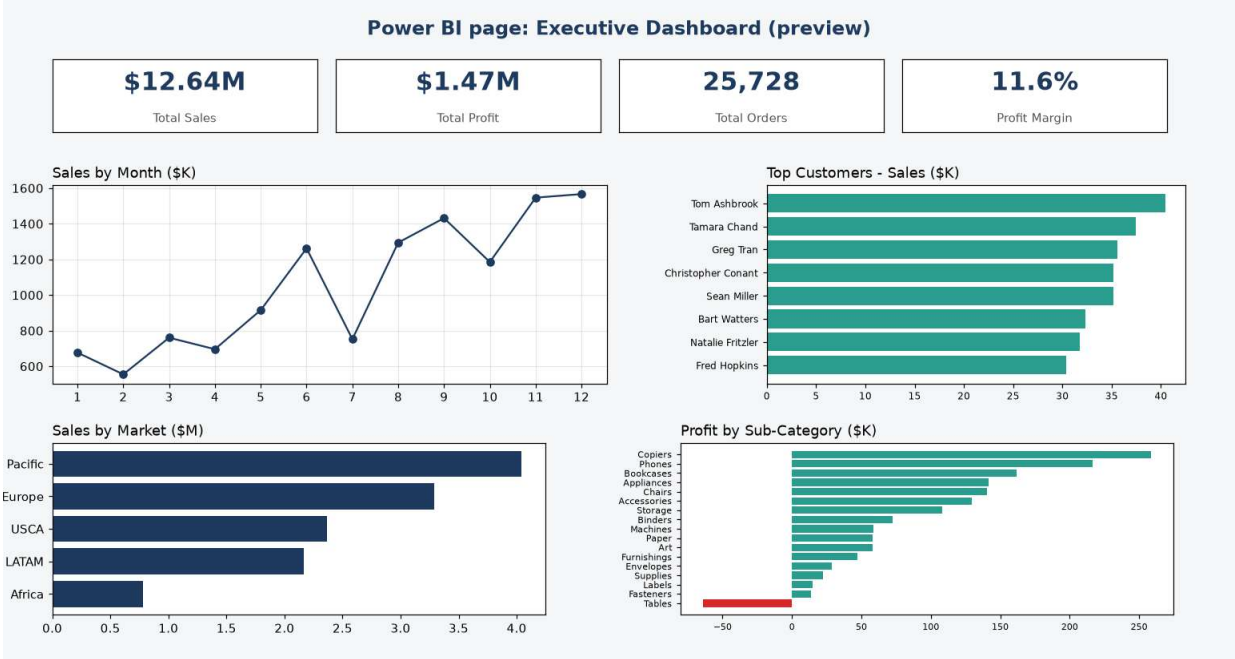


Figure 3: Executive Dashboard page (preview; replace with Power BI screenshot).

CONDUCT - PART B: WORKSPACE AND ACCESS PLANNING

Step 6: Workspace structure: three workspaces - Superstore-Dev (development), Superstore-Test (user acceptance testing) and Superstore-Prod (production, content shared to users through a Power BI App).

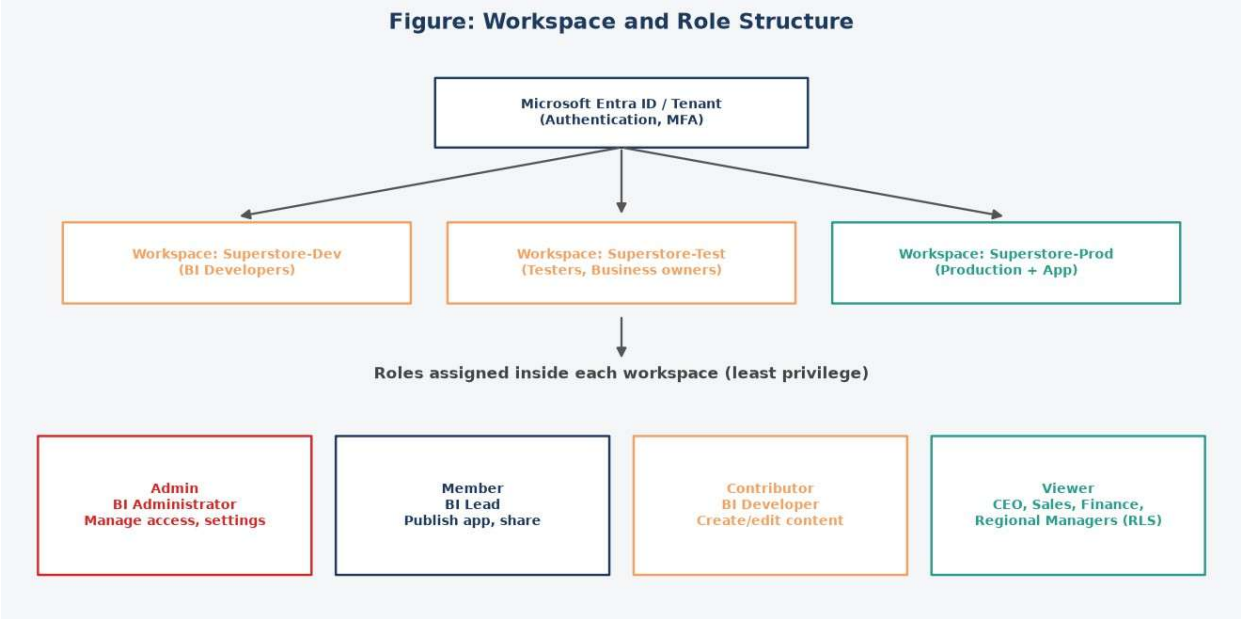


Figure 4: Workspace and role structure diagram.

Step 7-8: Roles and responsibilities

Role	Typical user	Responsibilities	Key permissions
Admin	BI Administrator	Manage workspace, add/remove users, security, settings, delete workspace	All permissions
Member	BI Lead	Publish and update the App, share content, manage items	Edit, publish, share
Contributor	BI Developer	Create and edit reports, datasets and dashboards; cannot manage access	Create/edit
Viewer	CEO, Sales, Finance, Regional Managers	View reports and dashboards, apply RLS, subscribe	View only

Step 9: Workspace governance guidelines

- Naming standard: Superstore-<Environment>; items named <Subject> - <Version>.
- Only the BI team has Admin/Member/Contributor roles; business users get Viewer access through the App.
- Use security groups, not individual users.
- Production changes only through the pipeline after testing and approval (change control).

- Sensitivity labels on confidential data; export and sharing restricted.
- Quarterly access review and removal of unused content.

CONDUCT - PART C: MONITORING STRATEGY DESIGN

Step 10-11: Monitoring requirements and Monitoring Matrix

Metric	What is checked	Tool	Frequency	Threshold / Alert	Responsible
Dataset refresh status	Success/failure, duration	Service > Dataset > Refresh history	Daily	Failure or >10 min -> email + Teams	BI Operations
Dashboard usage	Views, unique users per report	Usage metrics report	Weekly	Low usage <10 views/month -> review	BI Lead
Report performance	Page load time, slow visuals	Performance Analyzer (Desktop)	Per release + monthly	Load >5 s -> optimize	BI Developer
User activity	Sign-ins, exports, sharing, deletions	Microsoft 365 audit log / Admin portal	Daily	Unusual downloads or deletions	IT Security
Data quality	Row counts, nulls, totals vs source	Power Query checks / reconciliation	Each refresh	Mismatch >1% -> investigate	Data Steward
Capacity / gateway	Gateway health, capacity use	Admin portal	Weekly	Gateway offline -> immediate	IT Operations

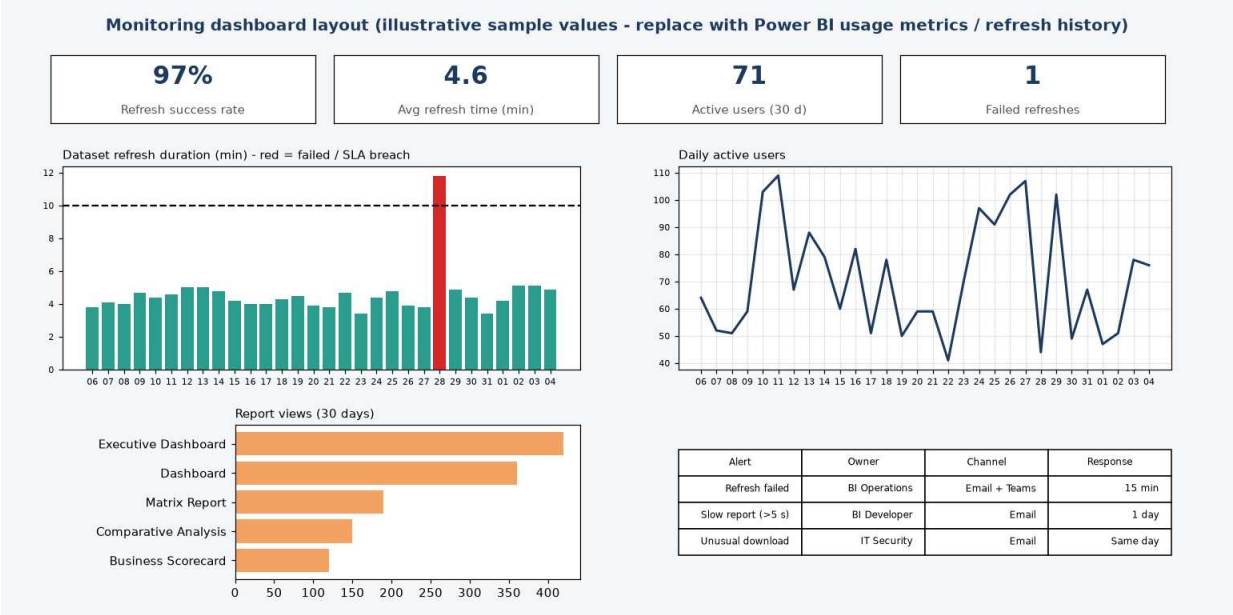


Figure 5: Monitoring dashboard layout (illustrative sample values; replace with real Power BI metrics).

CONDUCT - PART D: BACKUP AND RESTORE PLANNING

Step 12: Components requiring backup: datasets and source Excel file, report files (.pbix), dashboards and App definitions, security roles/permissions, scheduled refresh and subscription settings (documented), and the whole workspace archive.

Step 13: Backup Schedule (matches the table 'Backup Schedule' in the .pbix)

BI Component	Backup Frequency	Method	Storage	Retention	Responsible Person
Datasets and source Excel	Daily 23:00	Copy to versioned SharePoint / OneDrive	Primary + secondary location	30 days	BI Operations
Reports (.pbix)	Weekly (Sun 22:00)	Download from Service / Git or SharePoint version history	SharePoint + offline copy	12 weeks	BI Developer
Dashboards and App definition	Weekly (Sun 22:30)	Document layout, export PBIX/PDF snapshot	SharePoint	12 weeks	BI Lead
Security roles and	Weekly (Mon 06:00)	Export workspace	Restricted folder	1 year	BI Administrator

permissions		access list and RLS definitions			
Workspace archive	Monthly (1st)	Full export of all .pbix, settings document	Offline / archive storage	12 months	BI Administrator

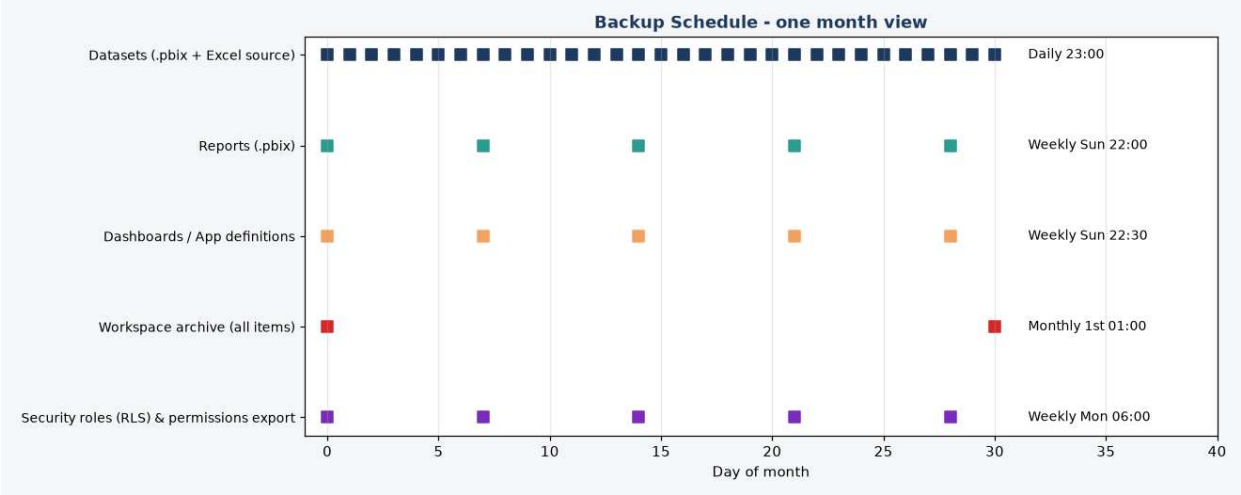


Figure 6: Backup schedule (one month).

Step 14: Restore procedure

- 1. Report the incident to BI Operations and note what was lost (item, time, who).
- 2. Check the workspace/admin recovery options first (deleted workspaces can be restored by an admin within the retention period).
- 3. Locate the latest backup (.pbix and dataset) in SharePoint version history.
- 4. Open the .pbix in Power BI Desktop, verify measures and visuals, refresh data.
- 5. Publish to the Production workspace (or promote from Test), re-link to the dataset.
- 6. Re-apply RLS role membership, scheduled refresh, subscriptions and alerts, and republish the App.
- 7. Validate totals against the source (Sales, Profit, Margin) and inform stakeholders.

CONDUCT - PART E: DISASTER RECOVERY SCENARIO

Step 15: Scenario - 'A critical executive dashboard is accidentally deleted one day before a management meeting.'

Step 16: How backup and restore recover the solution: the Executive Dashboard report is stored as a .pbix in the weekly backup and in SharePoint version history, and the dataset is backed up daily. The maximum data loss is therefore at most one day (RPO 24 h). The report is restored from the backup, republished to the production workspace and re-linked to the dataset, which

is refreshed to the current data. Because the pipeline and the documented settings exist, no redesign is needed.

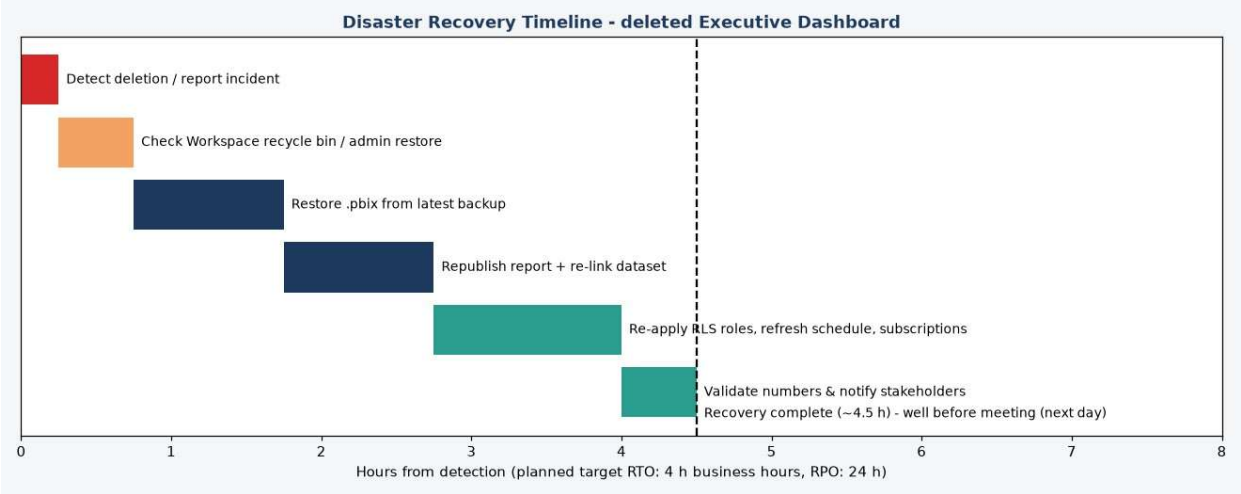


Figure 7: Disaster recovery timeline (target recovery within one working day).

Step 17: Actions for business continuity

Time	Action	Responsible
0 h	Detect and log incident; inform CFO/CEO office that the dashboard is being restored	BI Operations
0.5 h	Check recycle/admin restore options; identify latest backup	BI Administrator
1-2 h	Restore .pbix, republish to Production, re-link dataset	BI Developer
3-4 h	Re-apply RLS, refresh, subscriptions, App; validate numbers (Sales $12,642,502, Profit $1,467,457, Margin 11.6%)	BI Developer + Data Steward
Before meeting	Send a PDF snapshot of the dashboard to executives as fallback; confirm access	BI Lead
After meeting	Root-cause analysis; remove delete rights from non-admins; update procedure	BI Administrator

RISKS AND PREVENTIVE MEASURES

- Limit delete permissions to Admin/Member; give users Viewer access.
- Keep at least two backup locations and test a restore every quarter.

- Keep a PDF/PowerPoint snapshot of executive reports before important meetings.
- Document all settings so they can be re-created quickly.

DELIVERABLES

- BI Deployment Plan - Part A
- Workspace and Role Structure Diagram - Part B (Figure 4)
- Monitoring Matrix - Part C
- Backup and Recovery Plan - Part D
- Disaster Recovery Report - Part E

RESULT

A complete BI deployment, monitoring, backup, and recovery strategy was successfully designed. Deployment plans, monitoring procedures, workspace structures, and backup mechanisms were developed to ensure reliable and secure BI operations.

LEARNING OUTCOMES

- Analyzed deployment requirements for Business Intelligence solutions.
- Designed workspace structures and governance models for BI systems.
- Evaluated monitoring requirements for deployed BI solutions.
- Developed backup and recovery strategies for business continuity.
- Designed complete deployment and maintenance plans for organizational BI environments.